APPROXIMATING A MORPHISM OF QCQS SPACES BY A MORPHISM OF FINITE TYPE SYSTEMS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2806-2856, following the statement at 2795-2804.

Editorial note. No omissions; the proof consumes the published proposition-approximate as a prerequisite only.

Allowed standard prerequisites (editorial).

- Approximation by finite type spaces. proposition-approximate (same chapter, published as problem 1743 tag 07SU, cited at ~2807-2810). X = lim X_a and Y = lim Y_b with affine transitions.
- Fibre product criterion for relative limit preservation. lemma-better-characterize-relative-limit-preserving (same chapter, card L26304, cited at ~2814-2816 and ~2829-2831). Existence of f_{a,b} and the directedness lift a".
- Cofinal limits. Categories, Lemma categories-lemma-initial (cross-chapter, cited at ~2852-2854). Cofinal projections give lim_I X_i = X and lim_I Y_i = Y.

Source reference. Original reference 002R: focused-seeds/geometry/S9-spaces-limits-approximate-morphism/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0CP4: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1024–1024. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 07SU: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2328–2328. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of quasi-compact and quasi-separated algebraic spaces over $\mathbf{Z}$. Then there exists a direct set I and an inverse system $(f_i : X_i \to Y_i)$ of morphisms algebraic spaces over I, such that the transition morphisms $X_i \to X_{i'}$ and $Y_i \to Y_{i'}$ are affine, such that X_i and Y_i are quasi-separated and of finite type over $\mathbf{Z}$, and such that $(X \to Y) = \lim(X_i \to Y_i)$.*

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Write $X = \lim_{a \in A} X_a$ and $Y = \lim_{b \in B} Y_b$ as in Proposition 07SU, i.e., with X_a and Y_b quasi-separated and of finite type over $\mathbf{Z}$ and with affine transition morphisms.

Fix $b \in B$. By Lemma 0CP4 applied to Y_b and $X = \lim X_a$ over $\mathbf{Z}$ we find there exists an $a \in A$ and a morphism $f_{a,b} : X_a \to Y_b$ making the diagram

$$\begin{array}{ccc} X & \longrightarrow & Y \\ \downarrow & & \downarrow \\ X_a & \longrightarrow & Y_b \end{array}$$

commute. Let I be the set of triples $(a, b, f_{a,b})$ we obtain in this manner.

Let $(a, b, f_{a,b})$ and $(a', b', f_{a',b'})$ be in I. Let $b'' \leq \min(b, b')$. By Lemma 0CP4 again, there exists an $a'' \geq \max(a, a')$ such that the compositions $X_{a''} \to X_a \to Y_b \to Y_{b''}$ and $X_{a''} \to X_{a'} \to Y_{b'} \to Y_{b''}$ are equal. We endow I with the preorder

$$(a, b, f_{a,b}) \geq (a', b', f_{a',b'}) \Leftrightarrow a \geq a',\ b \geq b',\ \text{and } g_{b,b'} \circ f_{a,b} = f_{a',b'} \circ h_{a,a'}$$

where $h_{a,a'} : X_a \to X_{a'}$ and $g_{b,b'} : Y_b \to Y_{b'}$ are the transition morphisms. The remarks above show that I is directed and that the maps $I \to A$, $(a, b, f_{a,b}) \mapsto a$ and $I \to B$, $(a, b, f_{a,b})$ are cofinal. If for $i = (a, b, f_{a,b})$ we set $X_i = X_a$, $Y_i = Y_b$, and $f_i = f_{a,b}$, then we get an inverse system of morphisms over I and we have

$$\lim_{i \in I} X_i = \lim_{a \in A} X_a = X \quad \text{and} \quad \lim_{i \in I} S_i = \lim_{b \in B} Y_b = Y$$

by Categories, Lemma 002R (recall that limits over I are really limits over the opposite category associated to I and hence cofinal turns into initial). This finishes the proof. □